

Morphology

Class 9

Basic concepts

Overview: What is morphology?

- The study of the grammatical structure of *words*
 1. Processes of *word-making* (derivation: *run+er*)
and *word-marking* (inflection: *run+s*)
 2. The **internal structure** of **words**
 - their constituent parts and what they mean
 - how those parts can and can't be combined
 - Across the languages of the world
 3. People's **mental rules**
for constructing and understanding **words**
 - “rules” as consistent patterns, not prescriptive rules

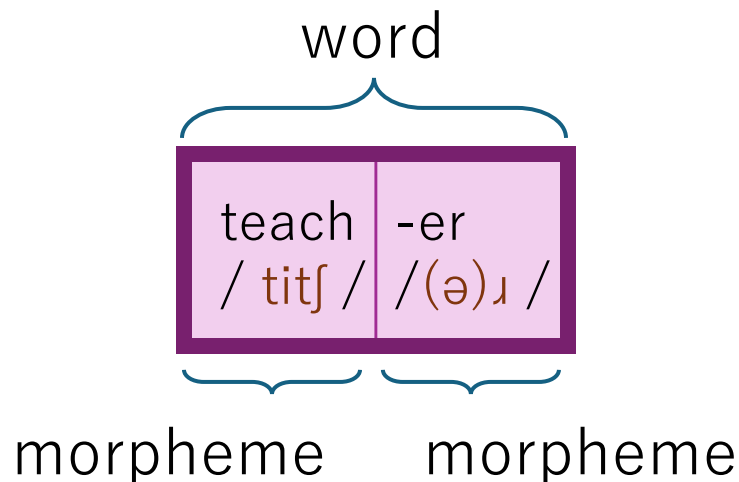
Overview: Important terminologies

• Word

- Smallest grammatically independent unit of language
- Assigned a particular part of speech (e.g., Noun, Verb)
- Made up of one or more **morphemes**

• Morpheme

- The smallest meaningful unit of language (irreducible)



Quiz 1: How many morphemes can you spot?

For each word below, decide how many morphemes it contains in Present-Day English.

unhappiness

☐ 1 ☐ 2 ☒ 3 ☐ 4

cats

☐ 1 ☒ 2 ☐ 3 ☐ 4

uncharacteristically

☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ 6

decentralization

☐ 2 ☐ 3 ☐ 4 ☒ 5 ☐ 6

Quiz 1: How many morphemes can you spot?

For each word below, decide **how many morphemes** it contains in Present-Day English.

reindustrialization

☐ 3 ☐ 4 ☒ 5 ☐ 6

indecisiveness

☐ 1 ☐ 2 ☐ 3 ☒ 4

counterrevolutionary

☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

unfriendliness

☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6

The citizens **revolted** against the dictator.

Quiz 1: How many morphemes can you spot?

Latin

volvere (volvō, volvī, volūtum)

- (i) vi. 'roll'
- (ii) vt. 'cause to roll'

re-volvere (revolvō, revolvī, revolūtum)

- (i) vi. 'roll back'
- (ii) vt. 'cause to roll back, revolve, unroll'

(1) *Pater* *libr-um* *re-vol-v-it*.

father.NOM book-ACC un-roll-**PRF**-3SG

‘The father has read (unrolled) the book.’

(2) *Liber* *re-vol-ūt-us* *est* (*ā* *patre*).

book.NOM un-roll-**PTCP.PR****F**-M is (by father.ABL)

‘The book has been read (unrolled) by the father.’

Quiz 1: How many morphemes can you spot?

Latin

volvere (volvō, volvī, volūtum)

(i) vi. ‘roll’

(ii) vt. ‘cause to roll’

re-volvere (revolvō, revolvī, revolūtum)

(i) vi. ‘roll back’

(ii) vt. ‘cause to roll back, revolve, unroll’

(3) *Crēdō* [*re-vol-ūt-iōn-em* *caelī*].

believe.1SG over-roll-PTCP.PRF-NMLZ-ACC.SG sky.GEN.SG

‘I believe in the revolution of the heavenly spheres.’

Etymologically the word ‘revolution’ can be split into 4 morphemes (in **Latin**).

However, in **English**, its compositionality is opaque; at least, “revolt” + “-tion” is what we are aware of.

Quiz 1: How many morphemes can you spot?

For each word below, decide **how many morphemes** it contains in Present-Day English.

reindustrialization

☐ 3 ☐ 4 ☒ 5 ☐ 6

indecisiveness

☐ 1 ☐ 2 ☐ 3 ☒ 4

counterrevolutionary

☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

unfriendliness

☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

Quiz 1: How many morphemes can you spot?

For each word below, decide how many morphemes it contains in Present-Day English.

English

receive

☒ 1 ☐ 2 ☐ 3 ☐ 4

deceive

☒ 1 ☐ 2 ☐ 3 ☐ 4

conceive

☒ 1 ☐ 2 ☐ 3 ☐ 4

perceive

☒ 1 ☐ 2 ☐ 3 ☐ 4

Latin *capere* 'to seize'

recipere (re-+capere)

☐ 1 ☒ 2 ☐ 3 ☐ 4

decipere (de-+capere)

☐ 1 ☒ 2 ☐ 3 ☐ 4

concipere (con-+capere)

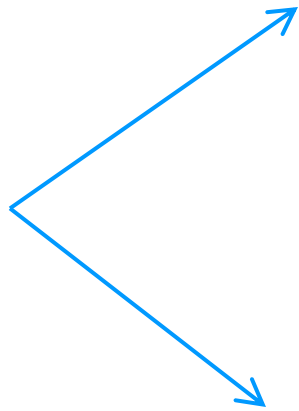
☐ 1 ☒ 2 ☐ 3 ☐ 4

percipere (per-+capere)

☐ 1 ☒ 2 ☐ 3 ☐ 4

capere 'to seize'

capere



receive
deceive
conceive
perceive

capture
captive
capacity
occupy
anticipate

Different types of morphemes

- **Free morpheme**

- A morpheme that can stand alone and be used as a word by itself

- **Lexical morphemes**

- *cat, dog, man, woman, perceive, ...*

- **Functional morphemes**

- *the, a, an, and, or, when, ...*

Different types of morphemes

- **Bound morpheme**

- A morpheme that must be attached to another morpheme

- **Derivational morphemes**

- Used to create **new words** and can change the lexical category of a word
- *pre-, un-, -ful, -ness, ...*

- **Inflectional morphemes**

- Used to **mark** a grammatical role in language
- Only 8 instances in English
- *-s (PL), -'s, -s (3sg), -ing, -ed, -en, -er (comp.), -est*

Quiz 2

Quiz 2 Terminologies

Which statement is correct?

- A. Every word consists of exactly one morpheme.
- B. Every free morpheme is a word, but not every morpheme is a word.
- C. Every bound morpheme is an inflectional morpheme.
- D. Every lexical morpheme is derivational.

B

- Free morphemes can stand alone as words (*book, run, happy*).
- Bound morphemes cannot (*un-*, *-ness*, *-s*).
- A is false because words such as **unhappiness** contain multiple morphemes.
- C is false because many bound morphemes are derivational (*un-*, *-ment*).
- D is false because lexical morphemes are content morphemes (*dog, sing, green*), not derivational affixes.

Quiz 3

Quiz 3: Terminologies

In the word *unfriendliness*, which classification is correct?

- A. *friend* is a functional morpheme.
- B. *un-* is a free derivational morpheme.
- C. *-ness* is a bound derivational morpheme.
- D. *-ly* is a bound inflectional morpheme.

C

- *friend*: free **lexical** morpheme
- *un-*: **bound** derivational morpheme
- *-ly*: bound derivational morpheme
- *-ness*: bound **derivational** morpheme

Quiz 4

Quiz 4: English tense markers

In English, the past tense marker **-ed** is typically analyzed as a **bound morpheme (suffix)**, as in (1)a.

- (1) a. Sarah **walk-ed** home. Past
- b. Sarah **will walk** home. Future

A student makes the following claim:

“*-ed* appears after the verb, so it is a **suffix** (-ed). *Will* appears before the verb. So, is it a **prefix** (*will-walk*), although orthographically we put a space in-between?”

- Do you agree with this proposal? Why or why not?

Quiz 4: English tense markers

In English, the past tense marker **-ed** is typically analyzed as a **bound morpheme (suffix)**, as in (1)a.

- (1) a. Sarah **walk-ed** home. Past
- b. Sarah **will walk** home. Future

No.

Data 1 Question test

- a. ***-Ed** Sarah walk home yesterday?
- b. **Will** Sarah walk home tomorrow?

Quiz 4: English tense markers

In English, the past tense marker **-ed** is typically analyzed as a **bound morpheme (suffix)**, as in (1)a.

- (1) a. Sarah **walk-ed** home. Past
- b. Sarah **will walk** home. Future

No.

Data 2 Adverb test

- a.* Sarah walk **probably** -ed home yesterday.
- b. Sarah will **probably** walk home tomorrow.

Class 10

Quiz 5

Quiz 5: English negation markers

In English, there are two ways to express **negation**:

(2) a. Sarah should **not** walk home.

b. Sarah should**n't** walk home.

The spelling may suggest that **n't** is a bound morpheme attached to **should**. **Do you think this analysis is appropriate? Why or why not?**

Yes.

Data 1 Adverb test

a. Sarah **should** **probably** **not** walk home.

b.* Sarah **should** **probably****n't** walk home.

Quiz 5: English negation markers

In English, there are two ways to express negation:

- (2) a. Sarah should **not** walk home.
b. Sarah should **n't** walk home.

The spelling may suggest that **n't** is a bound morpheme attached to **should**. **Do you think this analysis is appropriate? Why or why not?**

Yes.

Data 2 Question test

- a. **Should** Sarah **not** walk home?
b.* **Should** Sarah **n't** walk home?
c. **Should** **n't** Sarah walk home?

Quiz 6

Quiz 6: English comparative markers

English expresses comparison in two ways:

1. tall → taller
2. intelligent → more intelligent

What's the rule?

Quiz 6: English comparative markers

Adjective	Hint 1	Hint 2	Comparative
tall			tall er
short			short er
fast			fast er
happy			happi er
narrow			narrow er
simple			simpl er
clever			clever er / more clever
polite			more polite
famous			more famous
difficult			more difficult
beautiful			more beautiful
expensive			more expensive

N. of syllables

N. of syllables = 1 → **-er** , N. of S. = 2 → **-er/more** , N. of S. ≥ 3 → **more**

Excursion

What is a syllable?

Intuition: A **syllable** is a unit of pronunciation that typically contains **one vowel sound (or syllabic consonant)**.

Word	IPA	Number of syllables
cat	/kæ̃t/	1
happy	/hæ̃.pĩ/	2
beautiful	/bjũ.tə̃.fə̃l/	3
difficult	/dĩ.fĩ.kə̃lt/	3

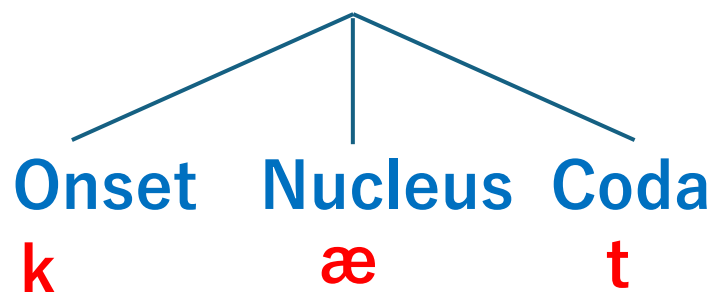
Excursion

Syllable structure

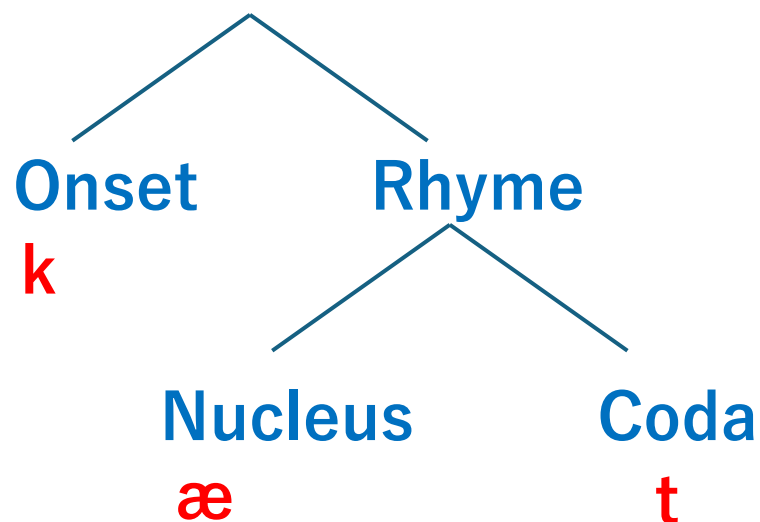
A syllable is often divided into three parts:

Word	IPA	Number of syllables
cat	/kæ̌t/	1
beautiful	/bju.ɫə̌.fə̌l/	3

*** Syllable**



✓ Syllable

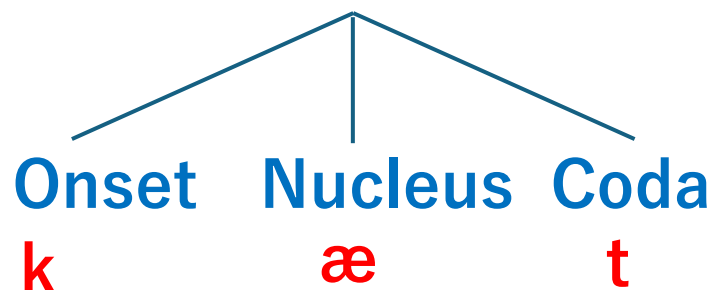


Excursion

Syllable structure

A syllable is often divided into three parts:

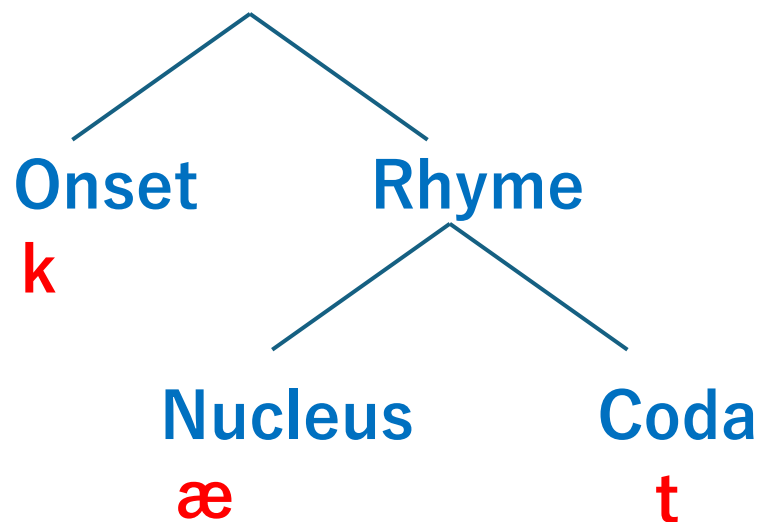
* Syllable



Evidence: Rhyming in poetry

cat – hat	Yes
cat – mat	Yes
cat – cap	No
cat – cut	No

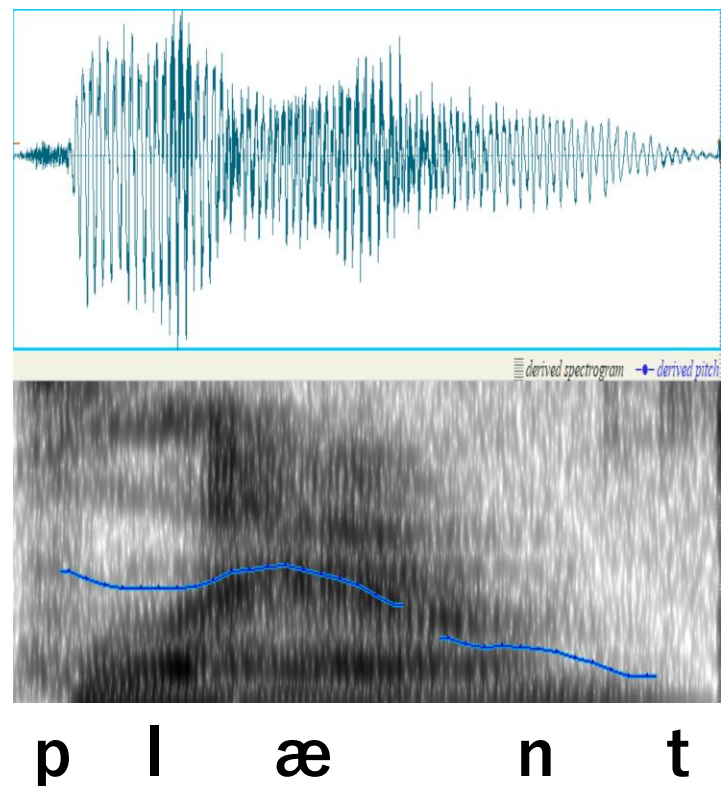
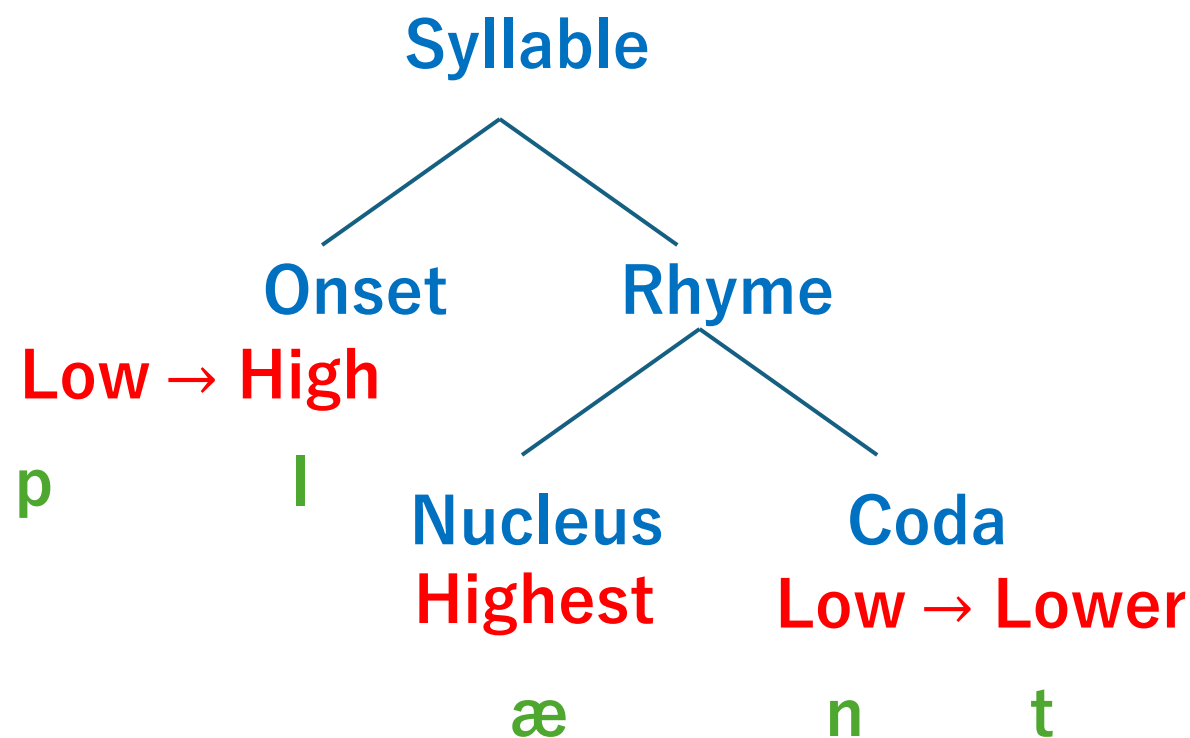
✓ Syllable



Excursion

Syllable structure

A syllable is often divided into three parts:



Quiz 6: English comparative markers

English expresses comparison in two ways:

1. tall → taller
2. intelligent → more intelligent

German also has a comparative suffix:

1. groß ("big") → größer ("bigger")
2. klein ("small") → kleiner ("smaller")
3. schön ("beautiful") → schöner ("more beautiful")
4. intelligent ("intelligent") → intelligenter
("more intelligent")

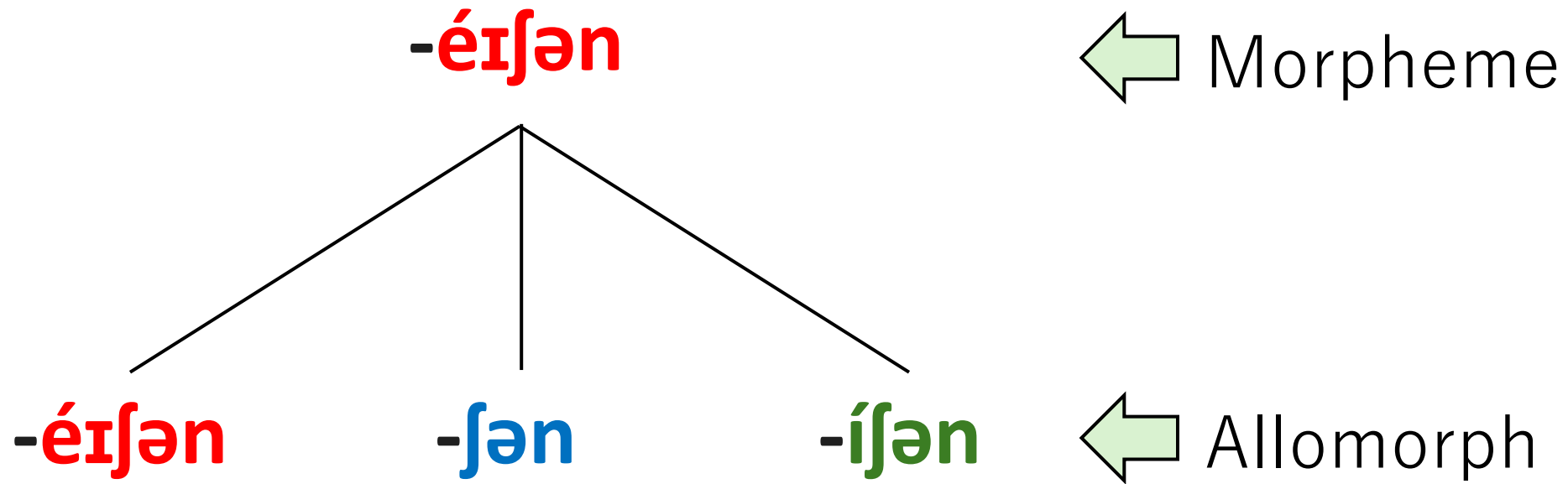
Class 10

Morphemes vs. Allomorph

Morpheme vs. allomorph

Verb	-ation	-(t)ion	-ition
réalizer /rí:əlaiz/	realizátion /rì:əlaizéɪfən/	*realiztion /rì:əlaizíjən/	*realition /rì:əl ífən/
éduquer /édʒʊkɛr/	*educatátion /èdʒʊkɛit éɪfən/	educátion /èdʒʊkéíjən/	*educition /èdʒʊk ífən/
repéter /rɛpɛ:t/	*repeatátion /rì:pi:t éɪfən/	*repeattion /rì:pi:tíjən/	repetítion /rèpət ífən/
commúne /kəmju:n/	*communátion /kòmɱjʊn éɪfən/	commúnion /kəmjuúnjən/	*communion /kòmɱjʊn ífən/
resúme /rizú:m/	*resumeátion /rì:zju:m éɪfən/	resúmption /rizámpjən/	*resumition /rì:zju:m ífən/
resólve /rizólv/	*resolvátion /rì:zɒlv éɪfən/	resolútion /rèzəlú:jən/	*resolution /rì:zɒlv ífən/

Morpheme vs. allomorph



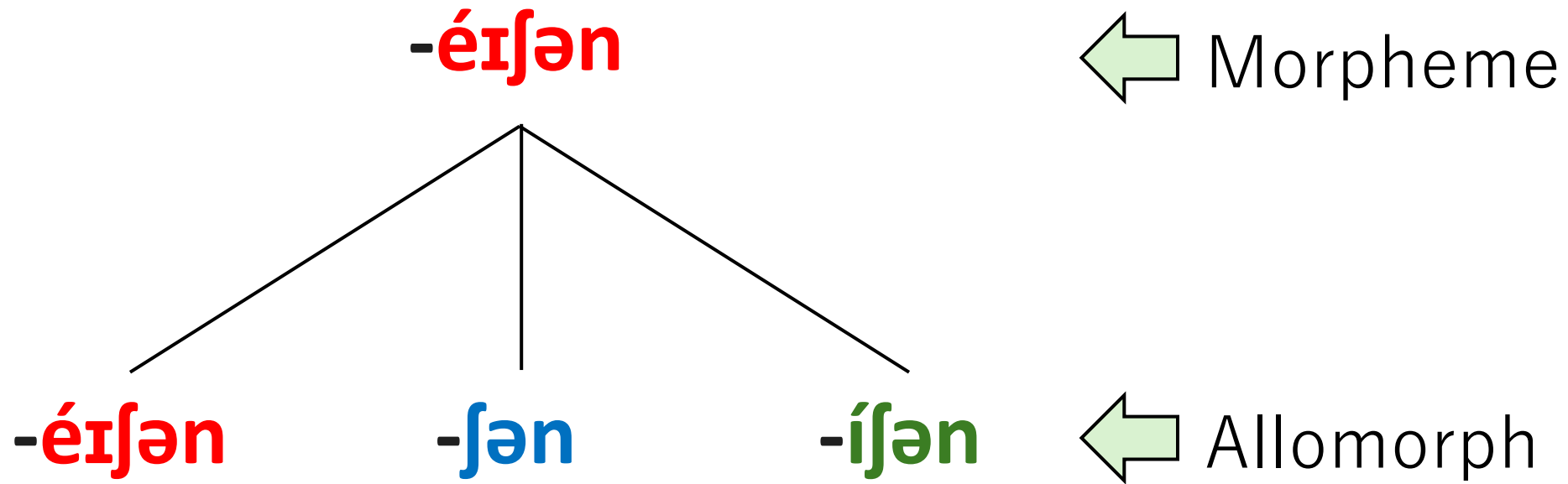
Morpheme

The smallest meaningful unit of language

Allomorph

A variant form of a morpheme that appears in a particular environment

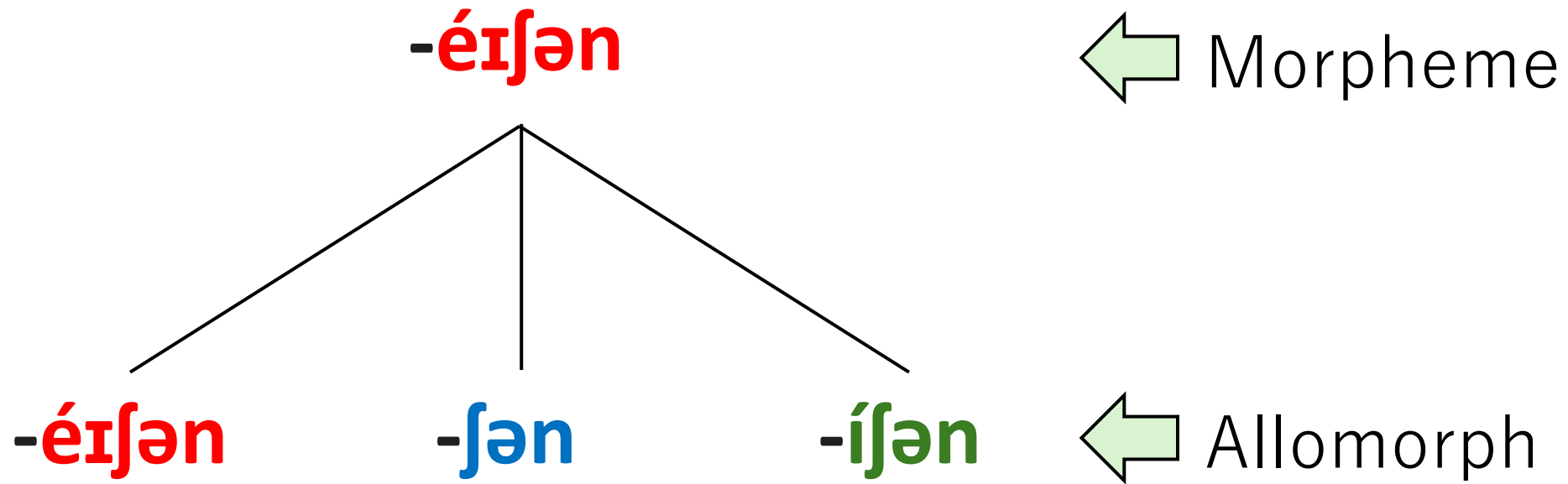
Morpheme vs. allomorph



Criteria for identifying allomorphs

- 1) **Meaning:** Semantically synonymous
- 2) **Distribution:** Complementary distribution.

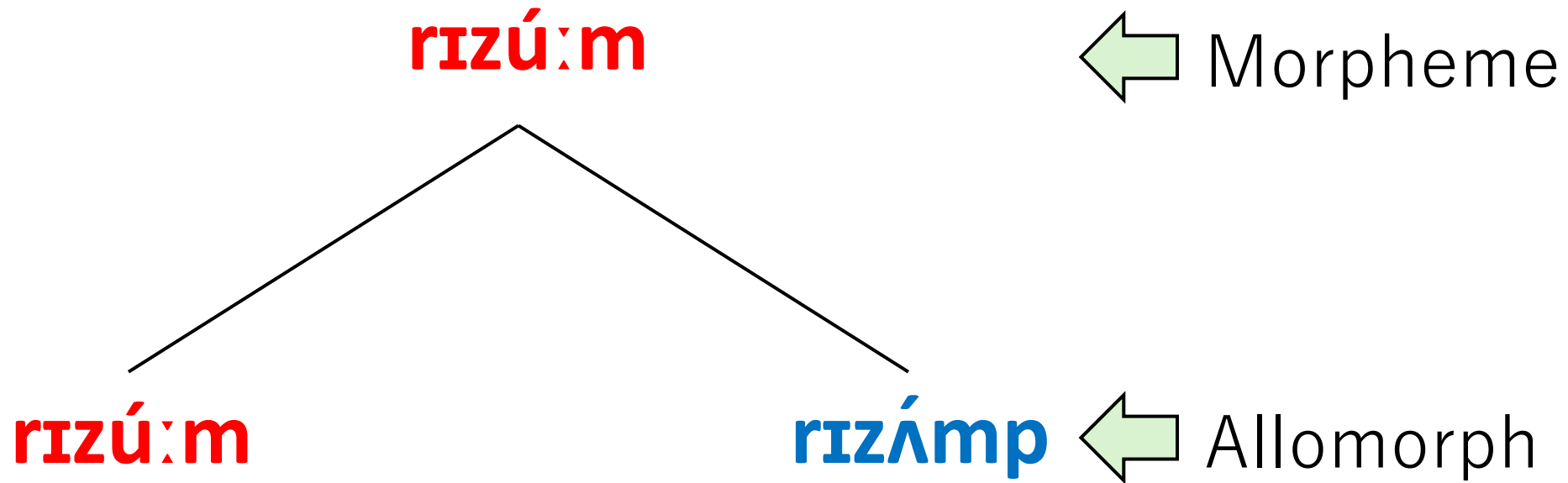
(Case 1) Affix allomorphy



Criteria for identifying allomorphs

- 1) **Meaning:** Semantically synonymous
- 2) **Distribution:** Complementary distribution.

(Case 2) Stem allomorphy



Criteria for identifying allomorphs

- 1) **Meaning:** Semantically synonymous
- 2) **Distribution:** Complementary distribution.

Morphemes vs. Allomorph

Quiz 7

Quiz 7: English derivational suffixes

When creating a noun from an adjective, English uses *-ity* or *-ness*. What are their differences?

1. human → human-**ity**
2. human → human-**ness**

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

-ity

yes

-ness

no

Data 1

Word-stress (phonology)

Adj

á.gile

fá.tal

pro.dú.c.tive

Adj-ity

a.gí.li.ty

fa.tá.li.ty

pro.duc.tí.vi.ty

Adj-ness

á.gi.le.ness

fá.tal.ness

pro.dú.c.ti.ve.ness

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

-ity

yes

limited

-ness

no

yes

Data 2

Productivity: nonce words

cringey-ness, Elon Musk-ness, Not-my-problem-ness, middle-of-the-road-ness,
...

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

-ity

yes

limited

limited

-ness

no

yes

yes

Data 3

Transparency in meaning (semantics)

1. human → human-**ity** (≠ property of being a human)
2. human → human-**ness** (property of being a human)

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

-al

yes

-ful

no

Data 1

Word-stress (phonology)

Adj

pa.**rén**.t

co.**l**or

e.v**én**t

Adj-al

pa.r**én**.tal

e.v**én**.tual

Adj-ful

co.**l**or.ful

e.v**én**t.ful

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

-al

yes

limited

-ful

no

yes

Data 2

Productivity: nonce words

Insta(gram)-ful, Disney-ful, Meme-ful, Goth-ful, ...

Ex. I need a slow, **coffee-ful** Sunday morning before this crazy week starts.

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

-al	yes	limited	limited
-ful	no	yes	yes

Data 3

Transparency in meaning (semantics)

1. event → event-**ual** (≠ property of being an event)
2. event → event-**ness** (property of being an event)

(Case 1) Derivation

Data 1

Word-stress
(phonology)

Data 2

Productivity

Data 3

Transparency
(semantics)

Data 4

Ordering

Class I affixes

-al yes
in-, -ity,
-ic, -ian

limited

limited

**inner
layer**

Class II affixes

-ful no
un-, -ness,
-less, -ish

yes

yes

**outer
layer**

(Case 1) Derivation

Data 4 Ordering

Class I affixes

-al

in-, -ity,

-ic, -ian

Class II affixes

-ful

un-, -ness,

-less, -ish

structure

$I < 2$

(Case 1) Derivation

Data 4 Ordering

Class I affixes

-al
in-, -ity
-ic, -ian

Class II affixes

-ful
un-, -ness,
-less, -ish

$I < 2$

color

$*2 < 1$

(Case 1) Derivation

Data 4 Ordering

Class I affixes

-al

in-, -ity,

-ic, -ian

Class II affixes

-ful

un-, -ness,

-less, -ish

$1 < 2$

$*2 < 1$

color **-ful** **-ness**

$2 < 2$

(Case 1) Derivation

Data 4 Ordering

Class I affixes

-al

in-, -ity,

-ic, -ian

Class II affixes

-ful

un-, -ness,

-less, -ish

1 < 2

*2 < 1

2 < 2

atom **-ic** **-ity**

1 < 1

(Case 1) Derivation

Data 4 Ordering

Class I affixes

-al
in-, -ity,
-ic, -ian

Class II affixes

-ful
un-, -ness,
-less, -ish

Level 1 Morphology
(Level 1 Phonology)

Level 2 Morphology
(Level 2 Phonology)

Nation-**al-iz-ation-al** -**less-ness**

1 < 1

1 < 2

*2 < 1

2 < 2

Quiz 8

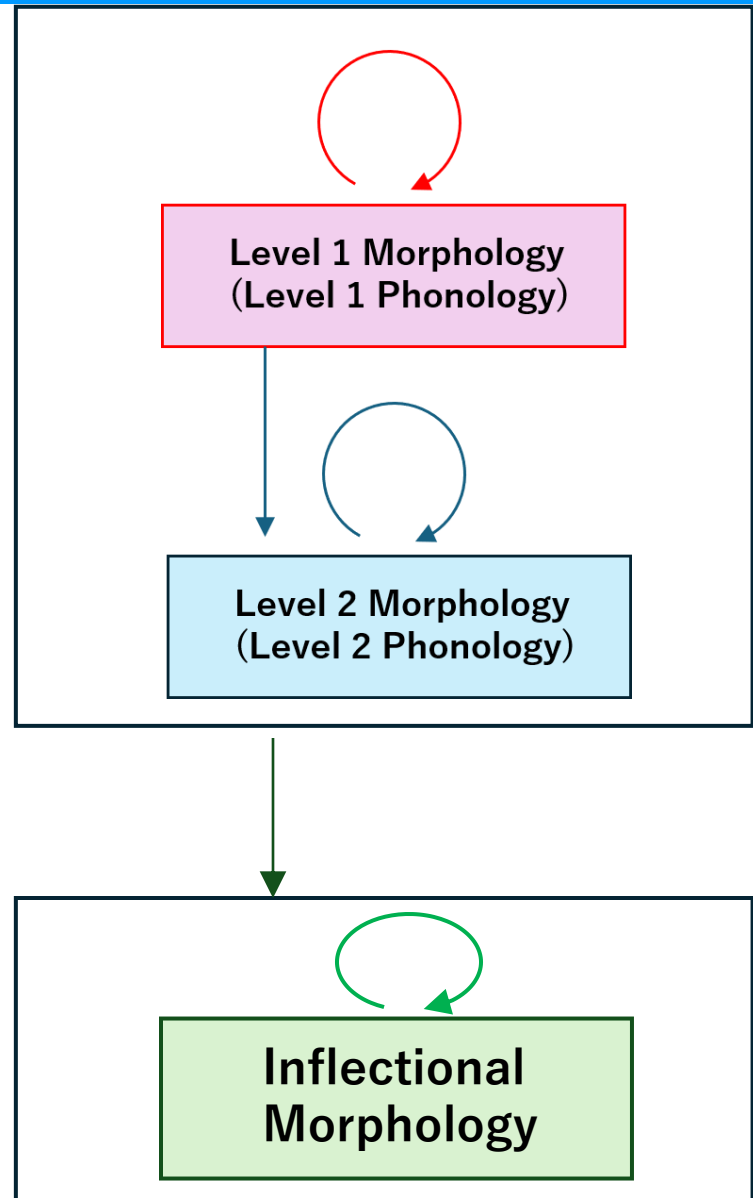
Quiz 8: Inflection

When an inflection is included in a noun, it must be taken away, even if it clearly suggests plural concepts. Why?

- (1) a.* brothers -hood
b. brother -hood

(Case 2) Inflection

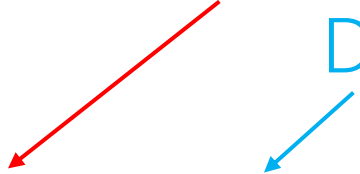
Class I affixes -al in-, -ity, -ic, -ian	Derivation (Word making process)
Class II affixes -ful un-, -ness, -less, -ish	
Inflectional affixes -ed, -en, -s, -ing, -er, -est, -'s	Inflection (Word marking process)



(Case 2) Inflection

Inflectional

Derivational

- (1) a.* brothers -hood
b. brother -hood
- 

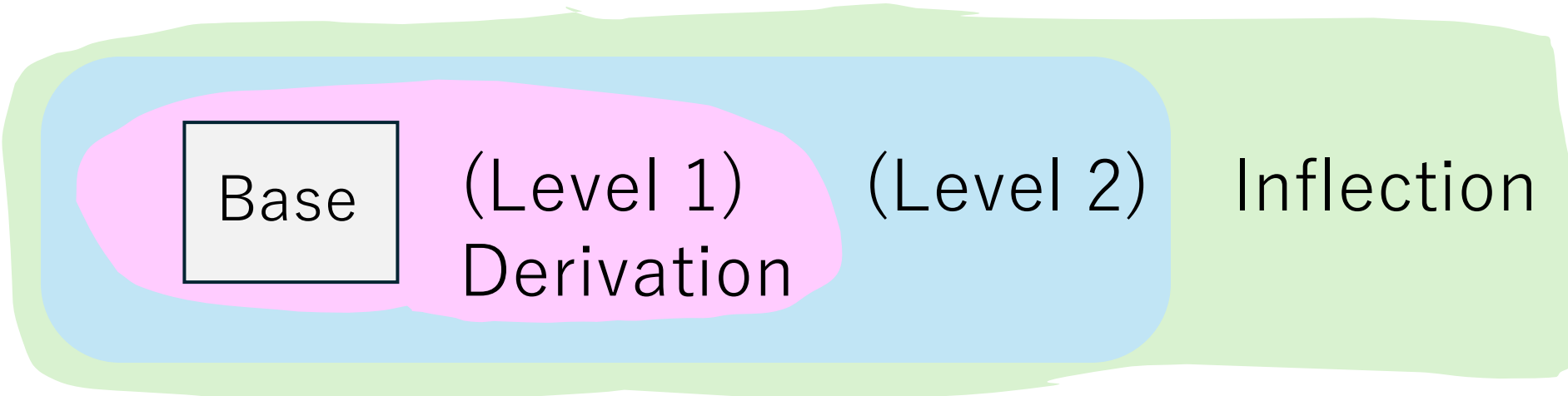
However,

- (2) a.* bird**s**-infested
b. **mice**-infested

-> Irregular processes are made at Level 1

Structure within a word

Hierarchical structure within a word



The diagram illustrates the hierarchical structure of a word using four nested, hand-drawn colored regions. From the innermost to the outermost, the regions are: a light gray square labeled 'Base', a pink oval labeled '(Level 1) Derivation', a light blue oval labeled '(Level 2)', and a light green oval labeled 'Inflection'. The regions are nested, with 'Base' inside the pink region, which is inside the blue region, which is inside the green region.

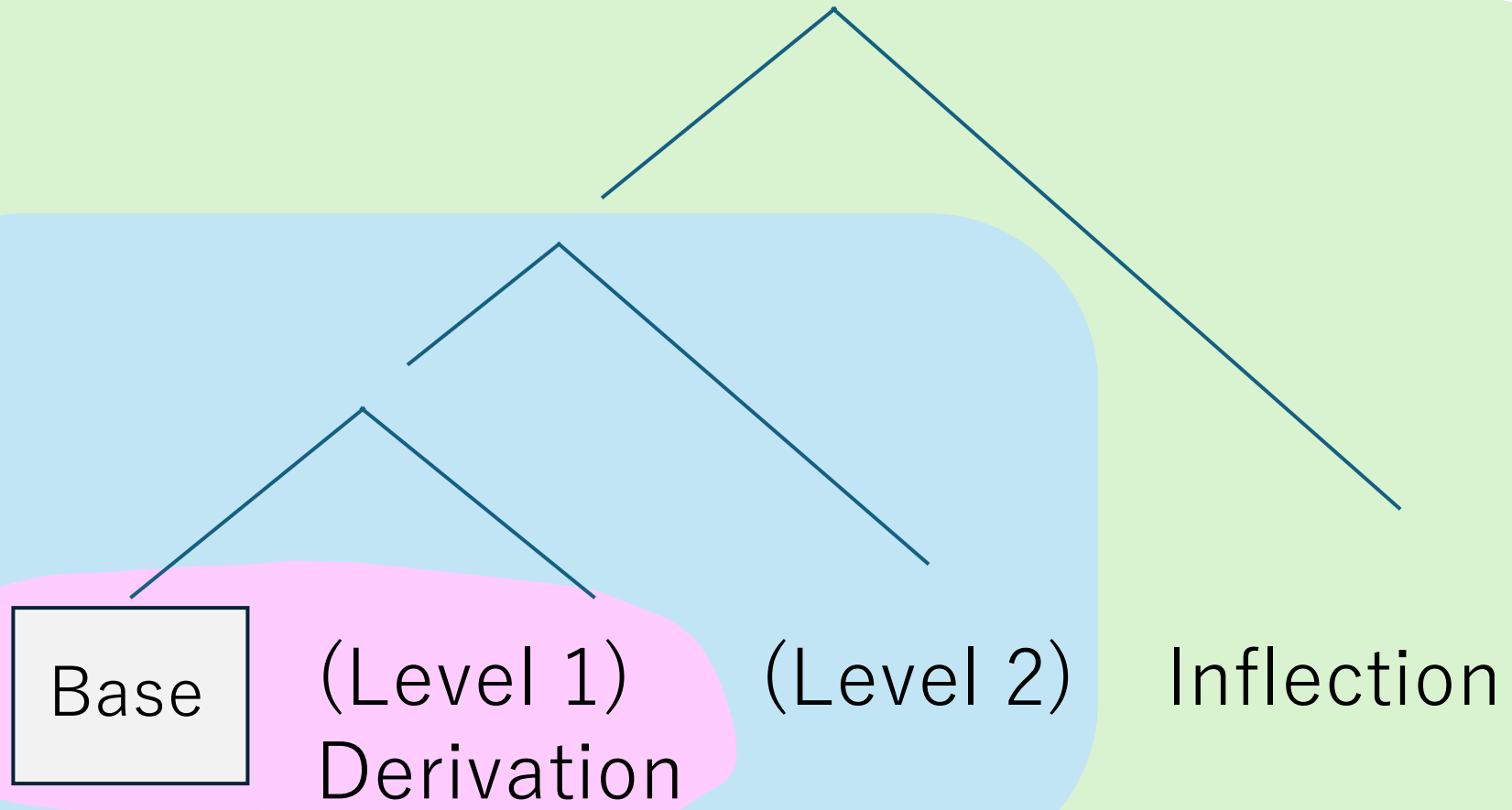
Base

(Level 1)
Derivation

(Level 2)

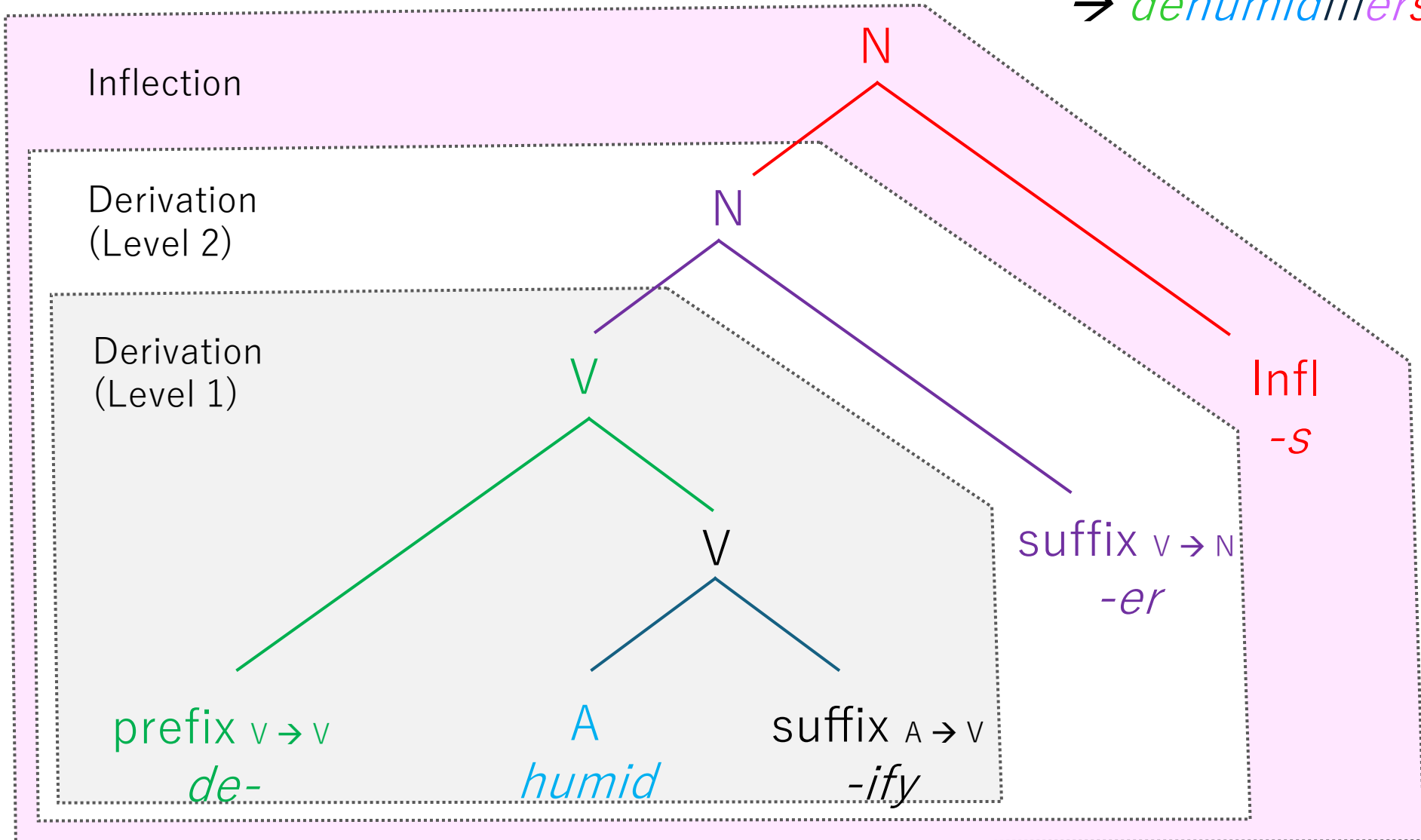
Inflection

Hierarchical structure within a word



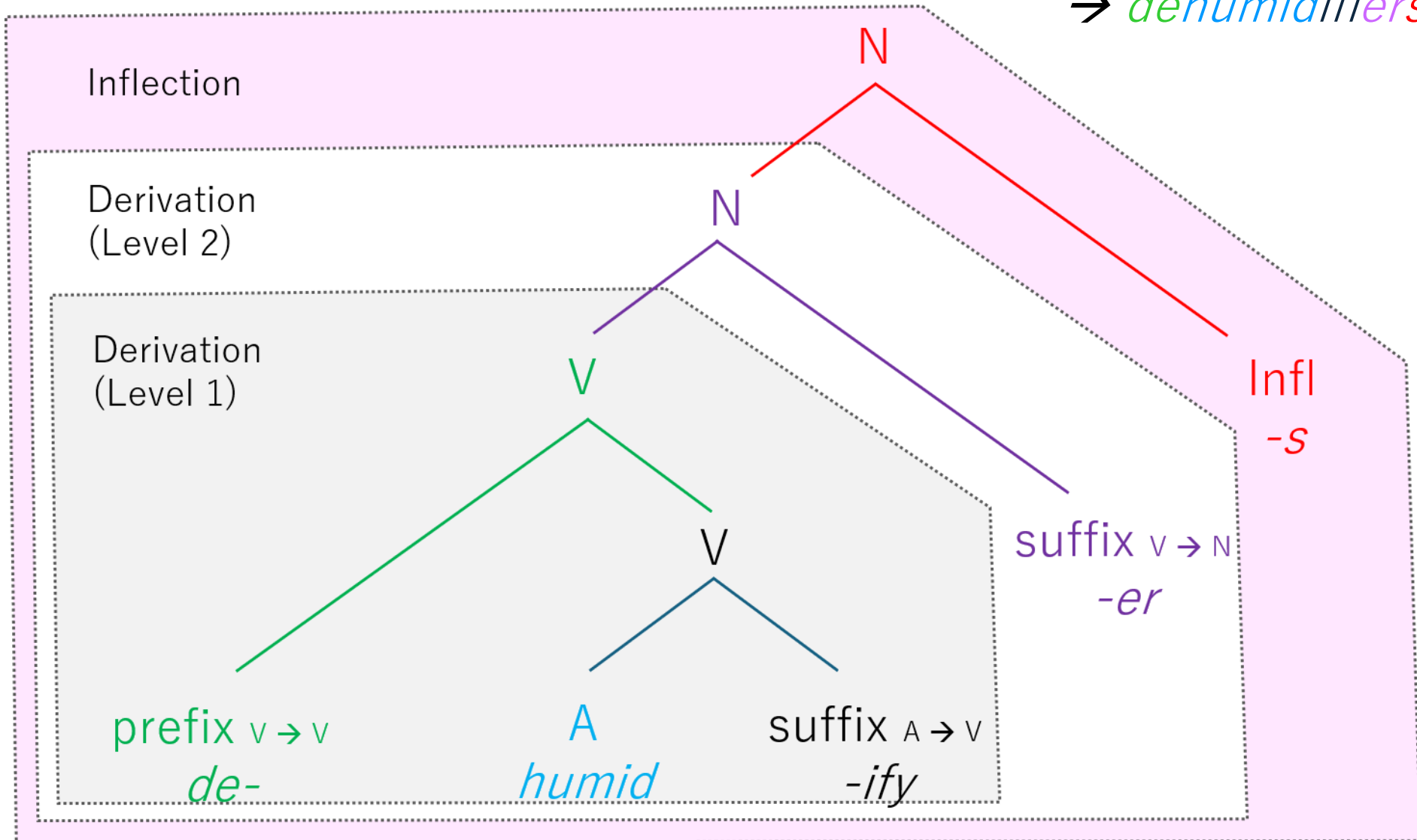
Hierarchical structure within a word

humid → *humidify* → *dehumidify* → *dehumidifier* → *dehumidifiers*



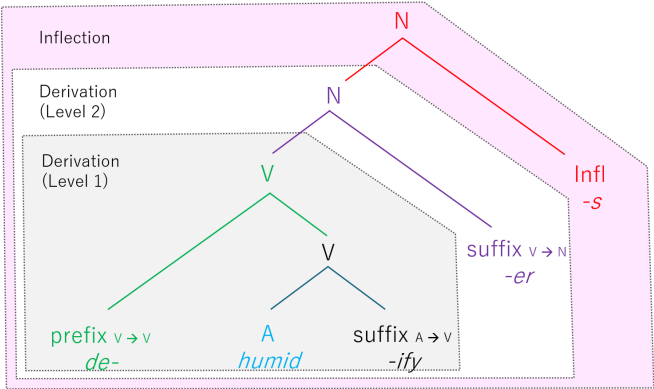
Hierarchical structure within a word

humid → *humidify* → *dehumidify* → *dehumidifier* → *dehumidifiers*



Hierarchical structure within a word

humid → *humidify* → *dehumidify* → *dehumidifier* → *dehumidifiers*



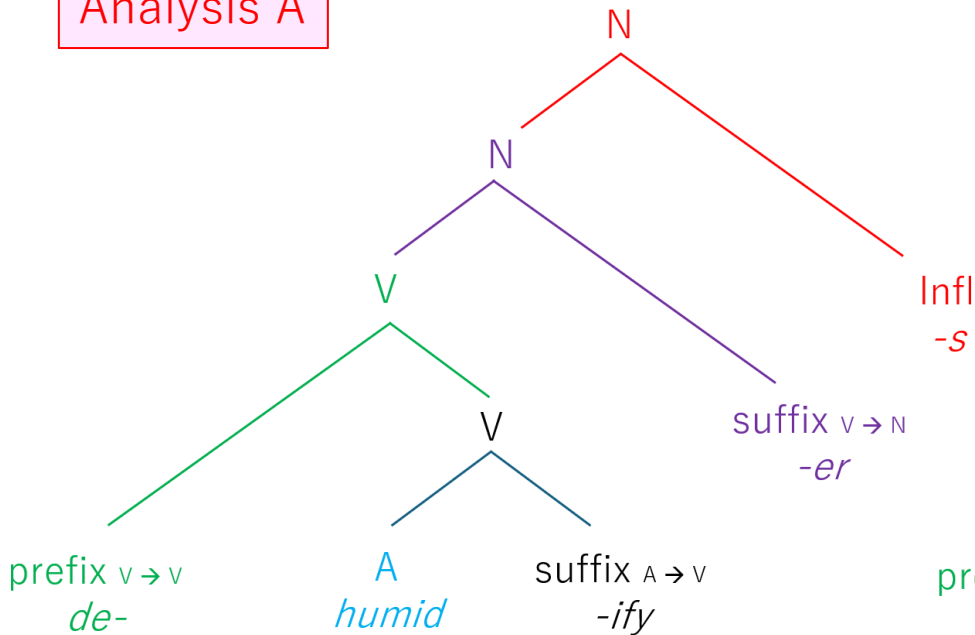
Morpheme	Type	Level	Origin	Selectional Restrictions (Base Type → Output Type)
de-	Prefix	Level 1	Latinate	Verb → Verb (Adds a privative/reversative meaning: "to reverse or remove")
humid	Root	Level 1	Latinate	Adjective (Serves as the free base for further derivation)
-ifi- (-ify)	Suffix	Level 1	Latinate	Adjective → Verb (Causative/inchoative meaning: "to make or cause to become")
-er	Suffix	Level 2	Germanic	Verb → Noun (Agentive/instrumental meaning: "an apparatus that performs the action")

Quiz 9

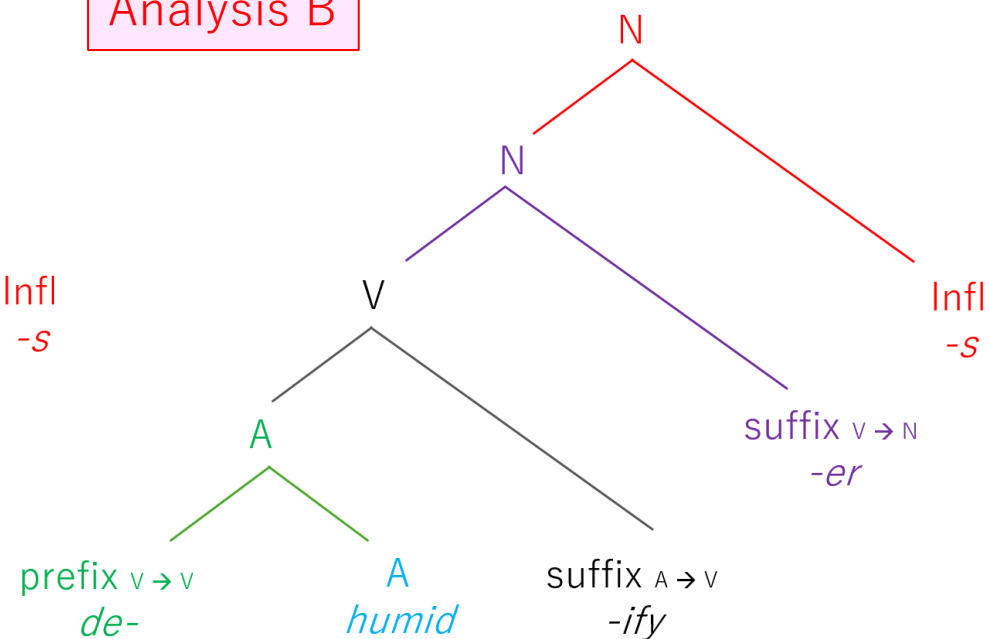
Quiz 9: Why is Analysis A better than Analysis B?

humid → *humidify* → *dehumidify* → *dehumidifier* → *dehumidifiers*

Analysis A

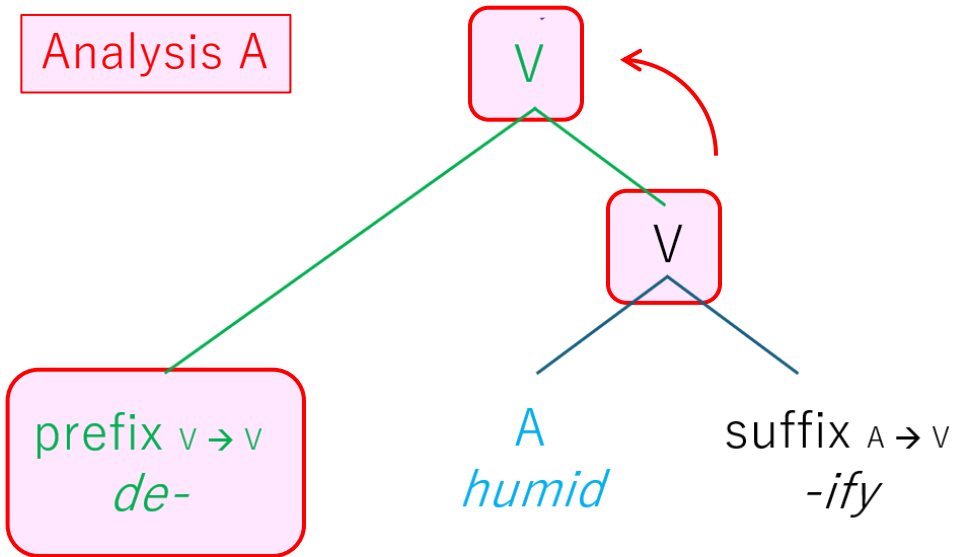


Analysis B

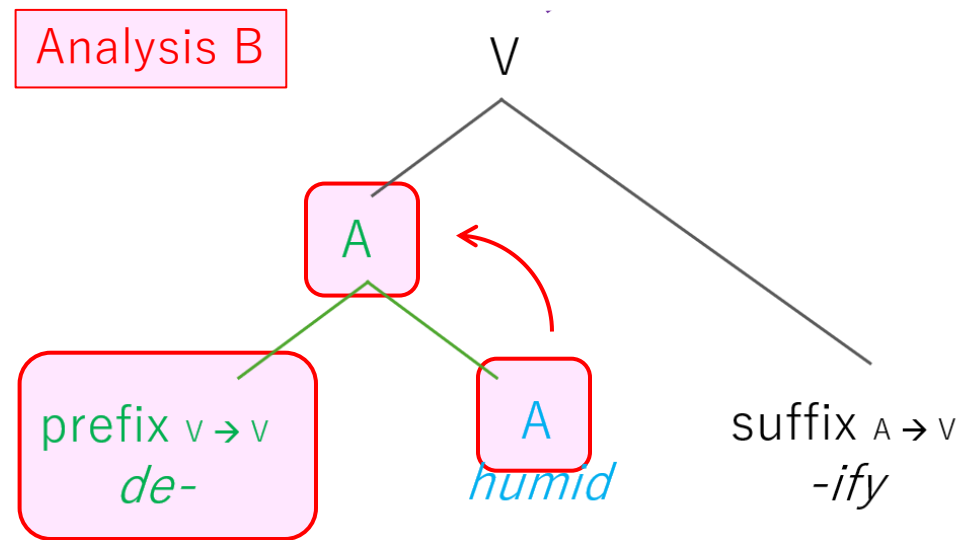


Selectional Restriction

Analysis A



Analysis B



Prediction A

Verb

Verb



compose
hydrate

→
→

de-compose
de-hydrate

Prediction B

Adj

Adj



honest
perfect

→
→

*de-honest (cf., dishonest)
*de-perfect (cf., imperfect)

Overview: Important terminology

- **Lexical morpheme (lexeme)**

- An root with a relatively rich, ‘vocabulary’-like meaning. The meaning allows us to describe the world – things, events, qualities of things and events, ...
 - Typically belongs to a ‘major’ part-of-speech category (noun, verb, adjective)
 - Often can serve as the root of a more complex word

- **Grammatical morpheme**

- A morpheme that contributes mainly grammatical information or has a meaning which is encoded as part of a grammatical paradigm
- For example:
 - English third-person singular *-s* (*eats*) simply agrees with the subject. No independent meaning.
 - English *of* in *a picture of Mary* links *picture* and *Mary*. The whole phrase indicates that Mary is the thing represented in the picture, but this specific relation is not the meaning of *of*.
 - English *the* indicates that something is familiar in in the discourse. It has meaning, but not the “world-describing” meaning of a lexeme.
 - Question particle *-ma* in Chinese.

- **Ambiguous cases (grammatical content morphemes)**

- Some morphemes are difficult to classify because they have world-describing semantic content but also contribute grammatical information:
 - English past tense *-ed* (*kissed*) brings in real world-describing meaning (the event took place in the past), but this meaning is part of a grammatical paradigm of past vs. nonpast.
 - English modal *can*

Practice

- Buffalo Bills players were reportedly impressed by the quality of toilet paper in the visitors' locker room at Lambeau Field Sunday, enthusiastically admiring the bathroom tissue's durability, absorbency, and softness.

What are the affixes and bound morphemes?

Practice

- Buffalo Bills players were reportedly impressed by the quality of toilet paper in the visitors' locker room at Lambeau Field Sunday, enthusiastically admiring the bathroom tissue's durability, absorbency, and softness.

*Where are the **affixes** and bound morphemes?*

Is that all?

Practice

- Buffalo Bills players were reportedly impressed by the quality of toilet paper in the visitors' locker room at Lambeau Field Sunday, enthusiastically admiring the bathroom tissue's durability, absorbency, and softness.

*Where are the **affixes** and bound morphemes?*

Is that all?

It's not always clear

- What do you think about:
 - *Sunday*
 - *Locker*
 - *Conceive*
- We have intuitions that these words have two parts, but are they based in...
 - History?
 - Grammar?
 - Psychological reality?

Morphological Operations

- **Affixation:** adding a morpheme to change or inflect a root
 - *Prefix:* a morpheme added to the beginning of a word

- Example:

English:

[in-] + human = inhuman ('not human')

PREFIX

ROOT

Chichewa:

Mkân̄go s- ú- na- ká-ngo-wá- phwányá
maûngu

3lion NEG-3SUBJ-PAST-go-just-6OBJ-smash 6pumpkins

'The lion did not just go smash them, the pumpkins'

Morphological Operations

- **Affixation:** adding a morpheme to change or inflect a root

- *Suffix:* a morpheme added to the end of a root

- Example:

English

reason + [-able] = reasonable

ROOT SUFFIX

Turkish:

Avrupa- li- laş- tir- il- a- mi- yacak- lar- dan-sin-iz

Europe-an-ize-CAUSE- PASSIVE- POTENTIAL- NEG-FUT- PART- PL- ABL-2nd-PL

‘You (all) are among those who will not be able to be caused to become like Europeans’

Morphological Operations

- **Affixation:** adding a morpheme to change or inflect a root

- *Circumfixes:* morphemes that are added around a root

- German past participle:

[ge]+zeig+[t]
PAST show PAST

*gezeig
*zeigt

- English:

[un]+[be]+know+[n]+[st]

*beknown
*knownst

Morphological Operations

- **Affixation:** adding a morpheme to change or inflect a root

- *Infixes:* morphemes that are inserted into a root

- Examples:

Tagalog:

tulong 'help'

t-um-ulong 'helped'

bili 'buy'

b-um-ili 'bought'

hanap 'search'

h-um-anap 'searched'

Morphological Operations

- **Root-and-pattern**

- Example: Semitic (Hebrew, Arabic) templatic morphology

- Arabic: 'ktb' = 'write'

katab 'to write'

kataba 'he wrote'

kutib 'has been written'

aktub 'be writing'

maktaba 'library, bookstore'

kitaab 'book'

kutub 'books'

kaatib 'clerk'

Morphological Operations

- **Root change:** a change on the {root, stem, base, lexeme} that indicates a change in grammatical function
 - Example: English irregular past (ablaut)
 - Fall → Fell
 - Give → Gave
 - Swear → Swore
 - Ride → Rode

Notice that it's pretty tough to think of a phonological rule that can account for this...

Morphological Operations

- **Suprasegmental Change:** changes like stress shift and tone shift that signal grammatical change
- Example: English stress shift:
 - Convíct (V) → Cónvict (N)
 - Progréss (V) → Prógress (N)
 - Rebél (V) → Rébel
 - How do we know that the noun comes from the verb, and not the other way?
 - Other nominal forms retain the verbal stress pattern but gain an affix:
 - Convíction, progréssion, rebéllion

Morphological Operations

- **Suprasegmental Change:** changes like stress shift and tone shift that signal grammatical change

- Example: Tone Shift in Somali

Singular

Plural

árday

ardáy

‘student(s)’

díbi

dibí

‘bull(s)’

mádax

madáx

‘head(s)’

túug

tuúg

‘thief(s)’

Morphological Operations

- **Reduplication:** grammatical change signaled by the repetition of some segment (a whole word, or just a syllable)
 - Example: Pangasinan (Austronesian, Philippines)
 - Too 'man' → to-too 'people'
 - Bii 'woman' → bi-bii 'people'
 - Plato 'plate' → pa-plato 'plates'
 - Baso 'glass' → ba-baso 'glasses'
- Do we have reduplication in English?

Identifying morphemes in an unfamiliar language

Source: *Language Files*, Volume 9

- Isolate the morphemes that correspond to the following English translations:

_____ possessive (genitive)

_____ 3rd person singular

_____ 2nd person plural

- List the allomorphs for the following translations:

_____ 'tortilla'

_____ 'chicken'

_____ 'rope'

- What are the phonological process conditions for these allomorphs?

1. [palu]	'stick'
2. [ku:ba]	'dough'
3. [tapa]	'four'
4. [geta]	'tortilla'
5. [bere]	'chicken'
6. [doʔo]	'rope'
7. [spalube]	'his stick'
8. [sku:babe]	'his dough'
9. [stapabe]	'his four'
10. [sketabe]	'his tortilla'
11. [sperebe]	'his chicken'
12. [stoʔobe]	'his rope'
13. [spalulu]	'your stick'
14. [sku:balu]	'your dough'
15. [stapalu]	'your four'
16. [sketalu]	'your tortilla'
17. [sperelu]	'your chicken'
18. [stoʔalu]	'your rope'

Two main types of morphology

- Derivational morphology: Word making
 - Intuitively, makes a new word.
- Inflectional morphology: Word marking
 - Intuitively, makes a new form of the same word.

Derivational morphology

- “Word making”
 - *active* → *activate*
- Often involves a change in meaning
 - May be more or less subtle
- Often (though not always) involves a change in lexical category (part of speech)
 - *love* → *loveable*
 - *happy* → *unhappy*
- Most commonly done through affixation
 - Zero-affixation: *love_V* → *love_N* “One who is easily the object of love”
 - Compounding: *greenhouse*, *bookkeeper*

Derivational morphology

- Creates new words from existing ones, allowing new words to enter a language

- Example: *dehumidifier*

dehumidifier

humid → *humidify* → *dehumidify* → *dehumidifier*

= ({ de_v [($humid_a$) ify_v] } er_n)

2 ROOT **1** **3**

= *humid*_{adj} + *-ify*_{verb} + *de-*_{verb} + *-er*_{noun}

Selectional restrictions on derivational morphemes

- Derivational affixes are typically **restricted** in what kinds of stems they combine with.
 - The derivational suffix *-able* attaches freely to verbs, giving you adjectives, but it doesn't attach to adjectives or nouns.
 - *adjustable, breakable, comparable, debatable*
 - **asleepable, *strongable, *studentable, *successable*
- In English, suffixes often change the category, but prefixes do not.

Selectional Restrictions

- Some affixes can attach to more than one lexical category.
Consider the derivational affix *pre-*:
 - [V → V]: *predetermine, premeditate, preboard*
 - [N → Adj]: *preseason, prewar, predawn*
 - [Adj → Adj]: *prefrontal, prehistoric, prenatal*

Compounding

- Compounding combines stems to form a new stems:
 - Don't assume that the use spaces, and hyphens tells you what's a word:
 - *fruit stand, blue-green, baseball*
 - Right-hand rule and exceptions
 - *fruit stand, blue-green*
 - *overweight, kickback*
- Notice that derivational affixation and compounding can happen in either order.
 - *man-eater*
 - *happiness quotient*

Some derivational affixes in English

Affix	Selected	Derived	Example
de-	V	V	decompress
dis-	V	V	disentangle
un-	Adj	Adj	unhappy
un-	V	V	unwrap
-able	V	Adj	washable
-ant	V	N	applicant
-ate	Adj	V	validate
-ful	N	Adj	beautiful
-ian	N	N	magician
-ify	Adj/N	V	purify, beautify
-ity	Adj	N	sensitivity
-ize	V	N	vaporize
-ment	V	N	management
-ness	Adj	N	happiness
-y	N	Adj	snowy

Inflectional morphology

- “Word marking”
- Adds **grammatical** and **semantic** information to a lexeme, in accordance with the particular syntactic requirements of a language
- Does not produce a change in lexical category

Inflectional morphology

- Indicates a property or a feature within an inflectional category. For example (there are many others):
 - Person
 - Number
 - Gender
 - Case
 - Tense
 - Aspect
 - Mood
- Think of examples of each from languages you know.

Inflectional morphology in English

- Nouns

- Plural
 - dogs**s**
- Possessive
 - John's**s**

- Adjectives

- Comparative
 - bigger**er**
- Superlative
 - biggest**est**

- Verbs

- 3rd singular present
 - runs**s**
- Past
 - dragg**ed**, ran
- Past participle
 - chosen**n**, broken**n**
- Present participle
 - walk**ing**, jump**ing**

Identifying morphemes in English

- Identify the morphemes
- Describe each as root, inflectional, or derivational

*In visions of the dark night
I have dreamed of joy departed –
But a waking dream of life and light
Hath left me broken-hearted.*

‘A Dream’, Edgar Allan Poe

Practice exercises

- The following are all from *Language Files*.

Identifying morphemes in English

- Divide the words below into their component morphemes. Decide for each morpheme if it is free or bound, what kind of affix is involved (if any), and (where applicable) if the affix is inflectional or derivational.
- Give a tree diagram showing the structure of each word.
- *Note:* words may consist of one, two, or more than two morphemes.

1. Comfortable
2. Massachusetts
3. environmentally
4. reconditioned
5. unidirectional
6. senseless
7. thickeners
8. nationalization
9. unspeakably

Suffixes

- In each group of four words below, two words have the same morphological structure, one has a different suffix from those two, and one has no suffix at all. Your task is to separate these groups according to the above information, tell which word(s) have the same suffix, which has a different suffix, and which has no suffix at all. Having done this, tell the meaning or function of each suffix.

- rider
- colder
- silver
- smoker

- tresses
- melodies
- Bess's
- guess

- running
- foundling
- handling
- fling
- tables
- lens
- witches
- calculates
- nicer
- painter
- runner
- feather

Suffixes and parts of speech

- From the examples given for each of the following suffixes, determine:
 - the part of speech of the word whose stem the suffix combines with
 - the part of speech of the words resulting from the addition of the suffix
1. –ify: solidify, intensify, purify, clarify, rarify
 2. –ity: rigidity, stupidity, hostility, intensity, responsibility
 3. –ize: unionize, terrorize, hospitalize, crystalize, magnetize
 4. –ive: repressive, active, disruptive, abusive, explosive
 5. –ion: invention, injection, narration, expression, pollution
 6. –less: nameless, penniless, useless, heartless, mindless

Root Allomorphy

- Identify any allomorphs that you uncover.

Example: malign/malignant [**məlain**]/[**məlign**-ənt]

- autumn/autumnal
- hymn/hymnal
- damn/damnation
- condemn/condemnation
- divide/divisible
- profane/profanity
- serene/serenity
- receive/receptive

Cebuano

The following nouns are from Cebuano, a language of the Philippine Islands. Examine them and answer the questions that follow.

- | | | |
|-----|-------------|------------------------|
| 1. | [bisaya] | ‘a Visayan’ |
| 2. | [inglis] | ‘an Englishman’ |
| 3. | [tagalog] | ‘a Tagalog person’ |
| 4. | [ilokano] | ‘an Ilocano’ |
| 5. | [sibwano] | ‘a Cebuano’ |
| 6. | [binisaya] | ‘the Visayan language’ |
| 7. | [ininglis] | ‘the English language’ |
| 8. | [tinagalog] | ‘the Tagalog language’ |
| 9. | [inilokano] | ‘the Ilocano language’ |
| 10. | [sinibwano] | ‘the Cebuano language’ |

- State the rule (in words, precisely) for deriving language names from the names of ethnic groups.
- What type of affixation is this?

Quiché

- _____ 'I'
 - _____ 'he'
 - _____ 'you'
 - _____ 'book'
 - _____ 'learn'
 - _____ 'read'
 - _____ 'the'
 - _____ 'weaving'
- _____ (present tense)
 - _____ (past tense)
 - _____ 'continually'
 - _____ 'yesterday'
 - What is the order of Quiché morphemes (in terms of subject, object, and tense marker)?

1. [kiŋsikíx le líbr] 'I read (present tense) the book'
2. [kusikíx le líbr] 'He reads the book'
3. [kiŋwetamáx le kém] 'I learn the (art of) weaving'
4. [kataxín kiŋwetamáx le kém] 'I continually learn the (art of) weaving'
5. [kataxín kawetamáx le kém] 'You continually learn the (art of) weaving'
6. [jiŋwetamáx] 'I learned (it)'
7. [juwetamáx le kém] 'He learned the (art of) weaving'
8. [jaxikíx le líbr iwír] 'You read the book yesterday'

Swahili

• [atanipenda]	‘s/he will like me’
• [atakupenda]	‘s/he will like you’
• [atampenda]	‘s/he will like him/her’
• [atatupenda]	‘s/he will like us’
• [atawapenda]	‘s/he will like them’
• [nitakupenda]	‘I will like you’
• [nitampenda]	‘I will like him/her’
• [utanipenda]	‘you will like me’
• [utmapenda]	‘you will like him/her’
• [tutampenda]	‘we will like him/her’
• [watampenda]	‘they will like him/her’
• [tulikulipa]	‘we paid you’
• [atanipiga]	‘s/he will beat me’
• [atakupiga]	‘s/he will beat you’
• [atampiga]	‘s/he will beat him/her’
• [ananipiga]	‘s/he is beating me’
• [anampiga]	‘s/he is beating him/her’
• [amekupiga]	‘s/he has beaten you’
• [amenipiga]	‘s/he has beaten me’
• [amempiga]	‘s/he has beaten him/her’
• [alinipiga]	‘s/he beat me’
• [alikipiga]	‘s/he beat you’
• [atakusumbua]	‘s/he will annoy you’

- _____ ‘I’
 - _____ ‘pay’
 - _____ ‘s/he’
 - _____ ‘me’
 - _____ ‘beat’
 - _____ ‘us’
 - _____ ‘we’
 - _____ ‘like’
 - _____ ‘annoy’
 - _____ ‘him/her’
 - _____ ‘they’
 - _____ ‘them’
 - _____ (past marker)
 - _____ (present progressive)
 - _____ (future marker)
 - _____ (present perfect)
 - _____ ‘you’ (if subject)
 - _____ ‘you’ (if object)
- What is the order of the morphemes in Swahili (in terms of subject, object, verb, and tense)?
 - Give the Swahili word for the following English translations:
 - ‘I have beaten them’
 - ‘They are beating me’
 - ‘They have annoyed me’
 - ‘You have beaten us’
 - ‘We beat them’
 - ‘I am paying him/her’
 - Give the English translation for the following Swahili words.
 - [atanilipa]
 - [utawapiga]
 - [walikupenda]
 - [nimemsumbua]